

SIMATS ENGINEERING

ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA1202	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Code of Conduct:

*I P Dinesh Reddy (192525399) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

ANNEXURE A INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization

A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

Smartphone Performance Optimization

Problem:

A smartphone manufacturer observes lag while switching between apps. The objective is to reduce memory access latency and improve overall throughput by optimizing the memory hierarchy.

Role of Memory Components

Memory	Role	Main Characteristic
L1 Cache	Stores frequently used instructions/data	Fastest, smallest
L2 Cache	Stores data not found in L1	Larger, slightly slower
L3 Cache	Shared cache for multiple CPU cores	Large, slower than L1/L2

Memory	Role	Main Characteristic
DRAM	Main memory for applications and OS	Large capacity, higher latency

Proposed Hierarchical Memory Design

CPU / Multiple Cores

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L1 Cache

Very Fast / Small

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L2 Cache

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L3 Cache

Shared / Larger

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DRAM

Large Main Memory

Optimization Approach

- Increase L1 cache efficiency for frequently accessed data.
- Optimize L2 cache to reduce accesses to slower memory.
- Use L3 cache for shared data between CPU cores.
- Use sufficient high-speed DRAM for active applications.
- Apply effective cache replacement and prefetching techniques.

Expected Result

The redesigned hierarchy reduces frequent DRAM accesses, thereby **reducing latency, minimizing app-switching lag, and improving system throughput.**

Conclusion:

A balanced **L1 → L2 → L3 → DRAM hierarchy** provides fast access to frequently used data while maintaining sufficient memory capacity, making it suitable for improving smartphone multitasking performance.

2. Cloud Server Memory Bottleneck

A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

Cloud Server Memory Bottleneck

Problem:

A cloud service provider experiences high latency in database queries. The objective is to compare **cache hierarchy and virtual memory paging** and recommend suitable memory technologies.

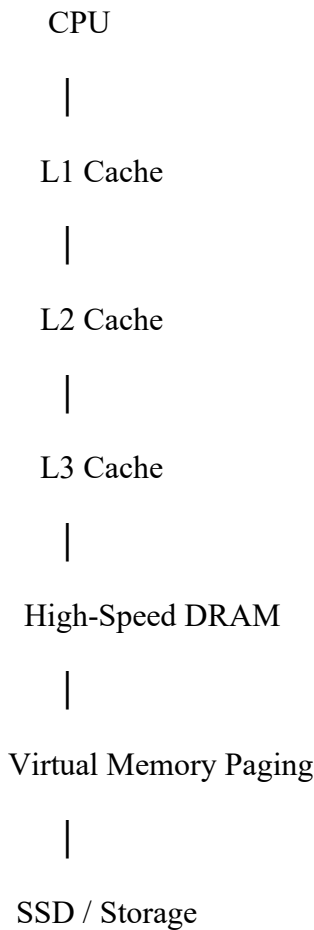
1. Cache Hierarchy vs Virtual Memory

Feature	Cache Hierarchy	Virtual Memory
Purpose	Reduces CPU memory-access latency	Extends available memory
Main Components	L1, L2, L3 cache	DRAM + storage
Speed	Very high	Lower
Access	Frequently used data	Data moved using pages
Effect on Queries	Faster data access	Prevents memory shortage

2. Recommended Solution

- Increase **L3 cache capacity** for frequently accessed database data.
- Use **high-speed DRAM** as the primary memory.
- Reduce excessive **paging** by providing sufficient DRAM.
- Use efficient **page replacement** to minimize page faults.
- Keep frequently accessed database pages in memory/cache.

3. Proposed Architecture



4. Expected Result

The combination of a **larger cache hierarchy and sufficient high-speed DRAM** reduces cache misses, page faults, and database access latency.

Conclusion:

For cloud database servers, **cache hierarchy should handle frequently accessed data**, while **DRAM should be increased to minimize virtual-memory paging**, resulting in lower latency and better throughput.

3. Autonomous Vehicle Data Processing

An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

Autonomous Vehicle Data Processing

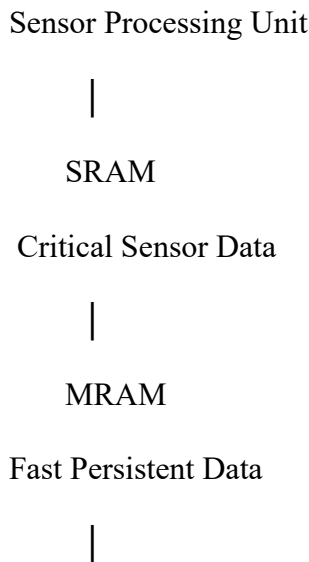
Problem:

An autonomous vehicle must process sensor data in real time. The memory system must maximize **bandwidth** and minimize **latency** using SRAM, DRAM, and MRAM.

1. Comparison of Memory Technologies

Memory Role	Key Characteristic
SRAM	CPU/cache memory for critical sensor data
DRAM	Stores large sensor datasets
MRAM	Stores frequently required data persistently

2. Proposed Memory Hierarchy



DRAM

Large Sensor Datasets

3. Design Approach

- Use **SRAM** for time-critical sensor processing.
- Use **MRAM** for frequently accessed persistent data.
- Use **DRAM** for large-volume camera, LiDAR, and radar data.
- Keep frequently required data closer to the processor to minimize latency.

4. Expected Result

The proposed hierarchy provides **fast access through SRAM**, persistent high-speed storage through **MRAM**, and sufficient capacity through **DRAM**.

Conclusion:

A combination of **SRAM + MRAM + DRAM** provides a practical memory hierarchy for autonomous vehicles, balancing **low latency, high bandwidth, and large data capacity**.

4. AI Inference Edge Device

An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

AI Inference Edge Device

Problem:

An edge AI device needs **fast model loading with low power consumption**. The task is to compare NAND Flash, DRAM, and RRAM and design an optimized hierarchical memory structure.

1. Comparison of Memory Technologies

Memory	Role	Key Characteristic
NAND Flash	Stores AI models permanently	High capacity, non-volatile
DRAM	Holds active models during inference	Fast access, volatile
RRAM	Stores frequently used AI model data	Low power, non-volatile

2. Proposed Memory Hierarchy

AI Processing Unit

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RRAM

Frequently Used Model Data

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DRAM

Active AI Models

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NAND Flash

Permanent Model Storage

3. Optimization Approach

- Store complete AI models in **NAND Flash**.
- Load active model data into **DRAM** for fast inference.
- Use **RRAM** for frequently accessed model parameters.
- Minimize unnecessary data transfers to reduce **power consumption**.
- Keep frequently used data closer to the processor.

4. Expected Result

The hierarchy provides **fast model loading, reduced memory-access latency, and lower power consumption** while maintaining sufficient storage capacity.

Conclusion:

A combination of **RRAM + DRAM + NAND Flash** provides an efficient memory hierarchy for edge AI devices by balancing **speed, capacity, persistence, and power efficiency**.

5. High-Performance Gaming Console

A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

High-Performance Gaming Console

Problem:

A gaming console suffers **frame drops during large scene loading**. The task is to analyze **L3 cache and DRAM throughput** and propose memory hierarchy enhancements.

1. L3 Cache vs DRAM

Feature L3 Cache

DRAM

Speed Very high

Lower than cache

Capacity Smaller

Much larger

Purpose Stores frequently used game data Stores large game assets

Effect Reduces CPU memory latency

Provides high-capacity data storage

2. Proposed Memory Hierarchy

CPU / GPU

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L1 Cache

|

L2 Cache

|

L3 Cache

Frequently Used Data

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High-Speed DRAM

Large Game Assets

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Storage

3. Enhancement Approach

- Increase **L3 cache capacity** to retain frequently accessed game data.
- Use **high-bandwidth DRAM** for large textures and scene data.
- Optimize data placement to reduce unnecessary DRAM accesses.
- Use efficient prefetching to load scene data before it is needed.

4. Expected Result

The improved hierarchy reduces memory-access delays and provides higher data throughput, helping to **minimize frame drops during large scene loading**.

Conclusion:

Combining a larger **L3 cache** with **high-bandwidth DRAM** provides a practical solution for improving gaming performance and maintaining smoother frame rates.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis